Experiment No. 8
Verification of Product Authenticity using a Blockchain DApp on the Sepolia Testnet

Aim

To connect a decentralized application (Pramaan) to the Ethereum Sepolia testnet through MetaMask and verify the authenticity and custody trail of a registered product using its on-chain product hash.

Theory

Decentralized application (DApp). A web front end that reads from and writes to smart contracts deployed on a blockchain. Here the front end is hosted on Vercel and the contract runs on the Sepolia testnet, so every record is public, tamper-resistant and verifiable by anyone.

Wallet connection. MetaMask injects an Ethereum provider into the browser. The site requests access to an account; the user approves with *Connect*, after which the DApp shows the connected address and network.

Product hash. Each product registered on the contract is identified by a 32-byte hash (`0x` + 64 hex characters). Verification means reading the stored record for this hash from the chain and checking its status and custody history.

Scan nonce. An optional `bytes32` value tied to one scan. It is used to detect replay of the same scan; if left empty, the DApp generates one automatically.

Terroir Score. An on-chain trust score (0–100) for a product. It starts from an AI photo-authenticity check done at registration and is adjusted as the product changes hands: it drops if an unverified handler joins the chain or if custody moves suspiciously fast.

Requirements

- Computer with a web browser and internet connection.
- MetaMask extension with an account on the Sepolia test network.
- Pramaan DApp (`https://pramaan-rho.vercel.app`) – *Verify* page.
- A registered product hash from the *Register Product* page.

Procedure

Step 1: Open the Verify page

Open the Pramaan DApp and go to the *Verify* tab.

Step 2: Connect MetaMask

The DApp requests a wallet connection on Sepolia. In the MetaMask pop-up select Account 1 and click *Connect* (Figure 1).

Step 3: Enter the product hash

Paste the product hash into the product code field. The scan nonce may be left blank, or a `bytes32` value can be entered.

Step 4: Verify

Click *Verify Product*. The DApp reads the verification result from the Sepolia contract (Figure 2).

Step 5: Check the result

The page shows the verification status and the Terroir Score of the product (Figure 3).

Output

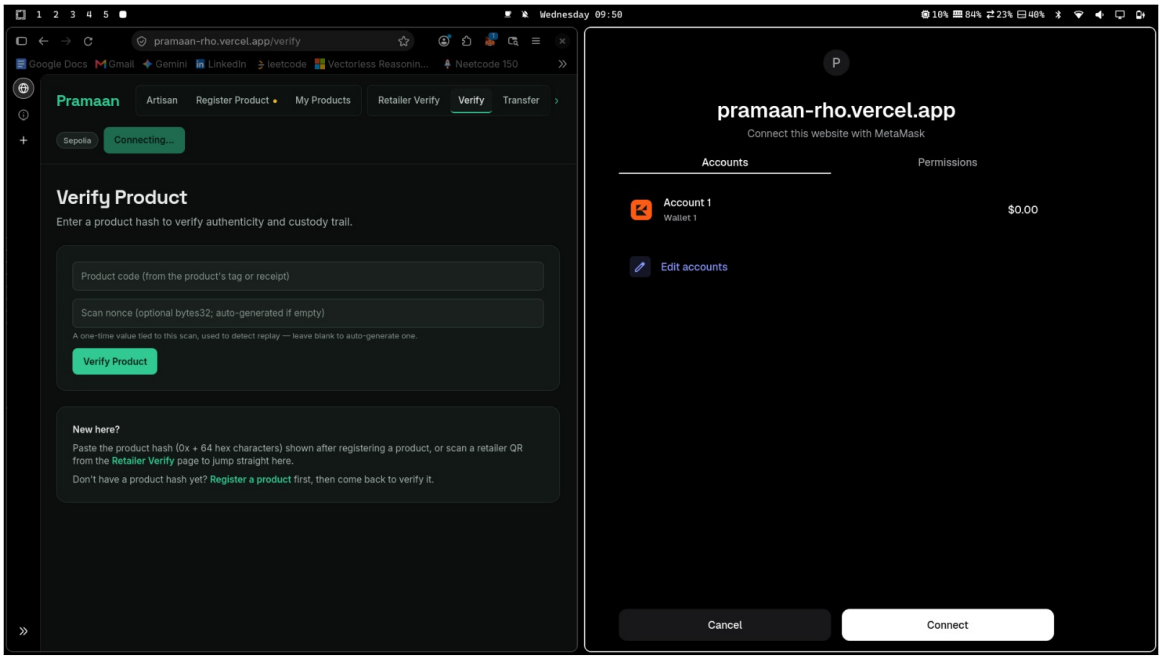


Figure 1: Connecting MetaMask (Account 1) to the Pramaan DApp on Sepolia

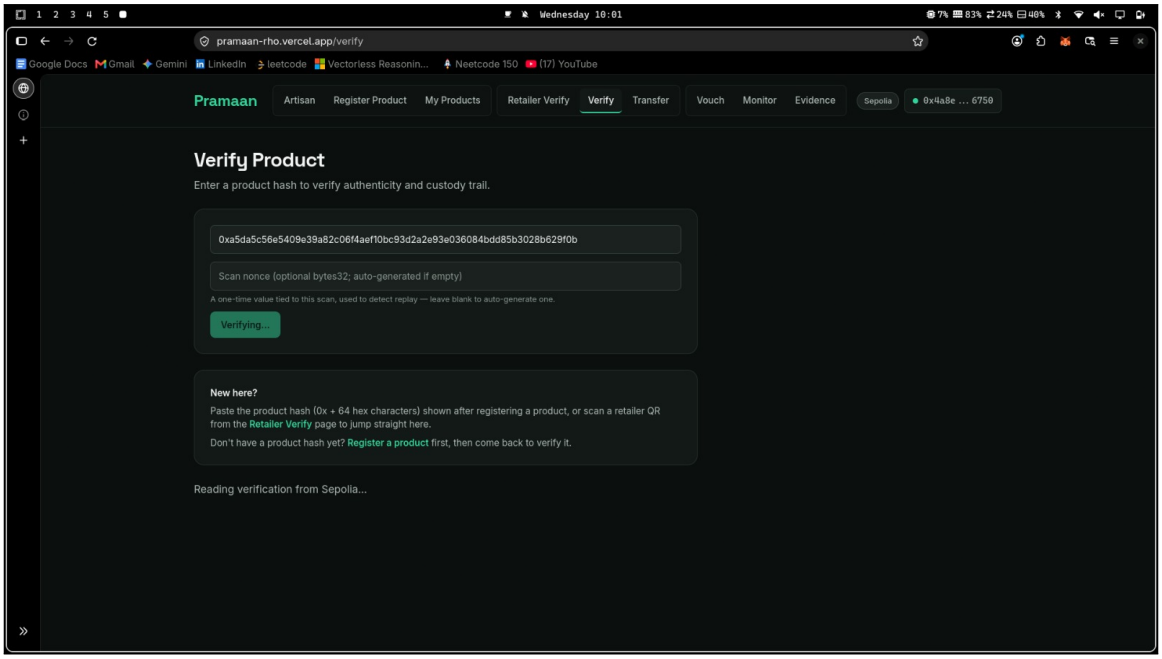


Figure 2: Verify page – product hash entered, reading verification from Sepolia

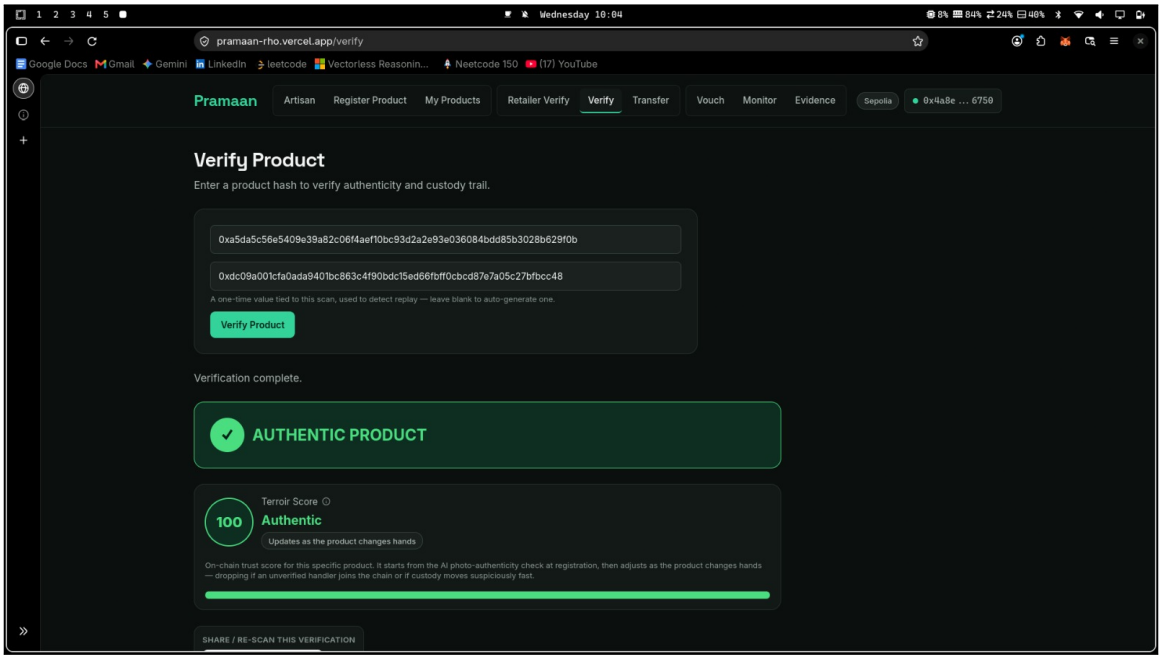


Figure 3: Verification complete – **Authentic Product**, Terroir Score 100

Conclusion

The Pramaan DApp was connected to the Sepolia testnet using MetaMask and a product was verified by its on-chain hash. The contract returned an *Authentic Product* result with a Terroir Score of 100, confirming that the product's record and custody trail are intact. This demonstrated how a blockchain can provide tamper-proof, publicly verifiable product authenticity checks.

Experiment completed successfully.